Multi-Stage Approach for Stress Detection Using Speech Lexical Analysis

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Phie Chyan
Department of Electrical Engineering
Hasanuddin University
Gowa, South Sulawesi, Indonesia
chyanp21d@student.unhas.ac.id

Andani Achmad
Department of Electrical Engineering
Hasanuddin University
Gowa, South Sulawesi, Indonesia
andani@unhas.ac.id

Ingrid Nurtanio
Department of Informatics
Hasanuddin University
Gowa, South Sulawesi, Indonesia
ingrid@unhas.ac.id

Intan Sari Areni
Department of Electrical Engineering
Hasanuddin University
Gowa, South Sulawesi, Indonesia
intan@unhas.ac.id

***Abstract*— Stress is one of the psychological problems that people most often experience. Stress that is not managed well and occurs for a prolonged time can lead to depression, negatively impacting a person's physical and mental health. Detecting and identifying stress is challenging, but it is necessary to provide appropriate early treatment for someone who experiences it. Many researchers have explored various technology-based approaches to detecting stress based on these needs. One approach currently being actively researched is stress detection technology using voice signals as an indicator to detect stress in a person. However, of these various studies, studies have yet to focus on exploring the use of voice properties optimally in terms of its properties as a signal wave and its output as a linguistic product. In this paper, we propose a multi-stage stress detection model through speech using a combination of speech signal characteristics and lexical aspects of speech. We employ the Mel-Spectrogram for the sound feature extraction required for stress detection from speech signals. Then, using Word2Vec and TF-IDF as input representations, we developed an NLP model for stress detection via lexical aspects of speech. The evaluation results of our proposed multi-stage stress detection model were able to detect stress well by showing an average accuracy and F1-Score of 91.7 and 91.2, respectively.**

Keywords*— *stress detection, speech signal, lexical analysis, mel-spectrogram, word2vec

I. Introduction

One of the psychological problems that people most often experience is stress, and stress that is not managed well and lasts for a prolonged time can cause depression [1]. This stressful event that leads to depression is a mental problem that cannot be taken lightly. Some symptoms related to depression are feelings of hopelessness, demotivation, withdrawal from social interactions, and mood swings, which occur repeatedly and cause the sufferer to experience psychological and physical damage depending on the severity of the problem [2], [3]. At a high level of severity, depression can affect brain activity and affect a person's behaviour, way of thinking and decision-making ability.

The problem of severe stress, which leads to depression, according to a survey conducted by WHO, has a fairly high prevalence globally. It is estimated that around 280 million people worldwide suffer from depression. The incidence of suicide, which is correlated with depression, is also quite high, with more than 800,000 cases reported worldwide every year [4]. One of the causes of the high number of cases of depression is the lack of availability of early treatment services, which means sufferers do not receive adequate treatment [5]. Some of the challenges faced include the subject's lack of awareness of the psychological problems they are experiencing and also the difficulty for the subject's close relatives to recognize these psychological problems [6]. A person's stressful state is generally reflected in their emotional expression and can be detected from their facial expressions and voice. Negative emotions such as anger, sadness, nervousness and fear that mark stressful situations can be indicators of psychological problems experienced by a person [7].

The general instrument used to diagnose stress is interview-based psychological assessments [8], [9]. This conventional method is unsuitable because it requires much patient background data. Besides that, this method requires honesty from the patient, where patients who experience psychological problems are generally vulnerable and try to hide or deny their psychological problems. Apart from interview-based methods, other methods rely on technology-based approaches using biometric body measurements such as heart rate, blood pressure, skin resistance, and other parameters [7], [10], [11]. Regarding the precision attained, this method is excellent in identifying stress. However, this method is not practical, as it needs various tools and sensors to identify different biometric traits. Often, this equipment can only be used in clinics or hospitals. In addition to these issues, using this procedure, which necessitates the attachment of numerous devices to the body, may be uncomfortable for certain patients, particularly those with allergies or skin sensitivity issues [12], [13].

Voice is a biomarker that can be used to detect stress because the sound output is a response from the brain's sympathetic and parasympathetic nerve circuits. Stressful episodes experienced by a person can affect the control of these nerves, which also control various muscles in the body and will trigger effects such as muscle tension, respiratory control and saliva production, which will affect sound reproduction [14], [15]. Changes in the body's working mechanisms when a person experiences physical and mental stress will eventually be translated into changes in various parameters in the voice, such as volume, pitch and speech prosody [16]. Apart from the vocal output, someone who experiences stress tends to communicate using language that contains negative expressions following the negative emotions they are experiencing. For example, they use

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sentences expressing sadness, anger, despair and various other negative expressions [17]. Based on the numerous parameters outlined, voice and lexical output can serve as effective biomarkers for stress identification. The ease of acquiring voice makes it possible to build a large dataset to support the stress detection model being built, which is one advantage of this method over baseline methods based on psychological assessments and biosignal parameters. Another advantage is that in terms of subject comfort, voice acquisition does not necessarily interfere with the subject's activities, preventing psychological effects like panic or nervousness before an examination.

In some recent research, the potential of voice as a promising biomarker in detecting stress has been explored [15], [18]. It is just that the use of sound features in these studies is more inclined towards extracting sound signal properties using artificial intelligence support to study signal patterns from the voice of someone experiencing stress, but the use of sound signal properties is generally not followed by analysis of lexical aspects as linguistic output. Whereas, as previously discussed, a person's state of stress can also be observed from what a person says because the words or sentences spoken reflect the feelings they are experiencing. Referring to the potential of voice, which is integrated with lexical analysis in stress detection modeling, in this research, we propose a multi-stage stress detection model through voice using signal characteristics and lexical aspects of speech.

The main contribution of this research is to propose a multi-stage stress detection model using signal features and lexical aspects of speech. The subject's voice will be captured by a microphone and sent into the model as input. The voice is recorded as a waveform, from which the signal domain feature will extract the sound features later. The results are then fed into a classification model based on a convolutional neural network that will temporarily predict the subject's stress status. Next, the model will analyze speech based on lexical features to identify potential stress in the subject's words or sentences. The final subject's stress status will be determined more precisely based on the findings of this proposed stress detection model.

In order to structure the presentation in this paper, the remainder of this paper is organized as follows: Section II discusses work related to this research topic. Section III explains the proposed methodology and model. Section IV documents the experiments and discussion based on the results obtained, and finally, Section V states the conclusions of this paper along with the continuation of future research directions.

II. Related Work

This section provides a broad explanation of the works relevant to this research's focus on numerous state-of-the-art studies on voice-based stress detection.

A. Stress Detection Methodology

First, Stress is a psychological issue with long-term negative repercussions on physical and mental health. It has spurred many studies to work on different methods of determining one's stress level. Nowadays, using interview-based psychological assessment tools is the gold standard for determining a person's level of psychological stress [15], [19]. Many standards, including the Patient Health Questionnaire and the Perceived Stress Scale (Morgan) (PHQ-8), are currently used. This interview-based examination is less efficient because it requires much time and money, and responders must meet specific cognitive aptitude criteria to take the test. Furthermore, this approach needs more validity and specificity [15].

Biological processes or biosignal-based measurement methods are another often utilized technique besides interview assessment-based techniques. This stress detection method offers advantages over interview-based assessment methods in preventing self-bias. In theory, this technique measures a variety of bodily reactions that stress-related stimuli cause. This biosignal-based detection method frequently uses heart rate, skin conductivity, blood pressure, and hormones like cortisol and adrenocorticotropics [20]–[22]. Depending on the type of parameter needed, various sensors or equipment are often employed and attached to the subject's body to get the necessary biosignal values. Currently, various more user-friendly devices are being developed, such as smartwatches that can monitor blood pressure and heart rate. However, in general, this biosignal measurement-based method is still relatively invasive, requiring blood or saliva samples from the subject [16].

The limitations of the previously discussed techniques increase the possibility of using alternative techniques for stress detection, such as stress detection through speech. Because they are inexpensive and non-intrusive, stress detection techniques based on speech are very promising. When speaking, a person prepares a string of words to be uttered in order to communicate with others and convey their message. Word choice, grammar, and word emphasis are just a few examples of how stressful situations can affect a person's use of language. As a result, language's lexical properties can be used as a stress indicator [23]. In addition to impacting language, stress also alters how the body functions, particularly the parts of the body that produce voice in humans, such as the vocal cords, muscles, and respiratory system [24]. Signal properties that detect various changes in a voice signal can be found by looking at the waveform. Typically, research that suggests a speechs-based stress detection model employs this technique by extracting features that can capture various alterations in the sound signal properties related to stressful situations.

B. Voice-Based Stress Detection Using a Machine Learning Approach

Deep learning is now widely used in many fields due to its ability to identify patterns in extensive data representations. This deep learning support has been applied in many research areas of image processing, computer vision, and signal processing [25], [26]. Research on voice-based stress detection also utilizes deep learning and machine learning. For example, research from [4] and [6] suggests a deep learning-based model that uses the CNN algorithm to detect subjects' stress status through voice. In extracting the features, the research utilized a mel-spectrogram and MFCC (Mel-frequency Cepstral Coefficients) extracted from the acquired sound waveform. These extracted features are then trained into a model to perform stress classification based on the characteristics of the subject's voice. Audio features in domain signals consist of time, frequency, and cepstral domains. Based on the audio feature analysis performed from those studies, it has been determined that the audio features of conversational sounds

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are more representative when extracted in the cepstral domain using audio features like Mel-spectrogram and MFCC. As a result, this feature is typically used in studies relating to analyzing and recognizing speech voice.

The voice acquisition process for stress detection is typically carried out in a controlled environment where noise or ambient sounds can be ignored or minimized [27]. The voice of each subject was recorded independently. Because the environment in which people interact is typically one that is filled with a lot of sound and noise, this is done to lessen the pre-processing load that must be applied to the acquired sound. The drawback of a procedure like this is that it makes the subject aware of the procedure being used, which leaves room for information bias. In addition to this issue, it is possible that by the time a person's voice is acquired, their moment of stress may have passed. Many studies have attempted to address this by conducting sound acquisition procedures directly in noisy environments, such as research from [27] and [28], but this method requires a more complex pre-processing method to separate noise, background sounds, and speech sounds so that the sound that goes into the feature extraction stage is entirely pure speech sound. The accuracy results of this procedure-based model are generally lower regardless of how sophisticated the pre-processing method is compared to models where the sound acquisition procedure is carried out in a controlled environment. This is because the noise that enters the feature extraction stage can affect model accuracy.

III. Proposed Methodology

The proposed model is a multi-stage stress detection model that performs hierarchical stress detection. The first stage of the model performs direct voice signal analysis through the voice waveform using low-level features related to the properties of the voice signal, and the second stage of the model performs speech recognition, converting it into text form and then analyzing the linguistic aspects using lexical features to predict the emotional sentiment of the sentence. The model then uses the multi-stage analysis results to predict the subject's stress status.

Fig. 1 demonstrates how the proposed stress detection model works. A microphone records the subject's voice and represents it as a waveform. The waveform is processed incrementally following the preprocessing stage. First, mel-spectrogram is used to extract sound features from the signal domain. After feature extraction, the feature vector obtained from the results is fed to a CNN classifier that is trained to identify stress in voice record. The CNN classifier's output will indicate whether the subject is experiencing stress and use this information for the next stage. According to the classification from the first stage, if the subject's stress status indicates a non-stress state, the model will declare that status as its final prediction status. However, suppose the CNN classifier detects a stress state. In that case, the model will continue stress detection in a subsequent stage by performing speech recognition and translation using a pre-trained Automatic Speech Recognition (ASR) model which will carry out automatic transcription (speech-to-text). The obtained text will be extracted using the Word embedding feature extraction model and statistical weighting based on Word2Vec + TF-IDF against the open-source dataset Dreaddit, a multi-domain text collection database for stress identification. The output of this lexical analysis model will provide a prediction of the subject's stress status based on sentiment analysis of the sentences spoken. If at this stage the model predicts the subject is in a stressful condition then the final prediction of the model is "stress". However, if at this stage the subject is predicted to be in a non-stressed state based on the context of their speeches, then the final prediction of the subject's stress status is "possible stress". A detailed explanation of each stage will be discussed further in this section.

A. First Stage: Stress Detection Through Speech Signal

In order to detect the stress using the properties of speech signals, the model in this study was trained using the open-source datasets Surrey Audio-Visual Expressed Emotion (SAVEE) and the Crowd-Sourced Emotional Model Actor Dataset (Crema-D). This dataset comprises 2800 and 480 voice sample files from various subjects, each representing a different emotion: happiness, pleasant surprises, sadness, anger, disgust, fear, and neutral. Based on psychological theory, negative emotional characteristics such as anger, sadness, disgust, and fear are characteristic emotions often experienced by someone who experiences stress. In contrast, someone who does not experience stress tends to show neutral emotions or joy and happiness [7]. Because in this study, the model only states a person's emotional status in a stressed or non-stressed state, we adjusted by relabeling the dataset. Voice samples that represent negative emotions like anger, sadness, disgust and fear are relabeled with the label "Stress," Conversely, voice samples that were originally labeled as neutral, happy, and pleasant surprises were relabeled as "Non-Stress." After relabelling, we obtained 1840 voice sample files labeled "Stress" and 1440 voice sample files labeled "Non-Stress."

After the relabeling process, we then carry out the data augmentation. Data augmentation is applied to increase the dataset's diversity so that the model has better generalization capabilities over various small perturbations in sound samples. The data augmentation process creates variations of new synthetic data samples based on voice samples from the dataset by injecting artificial noise, modifying pitch, and

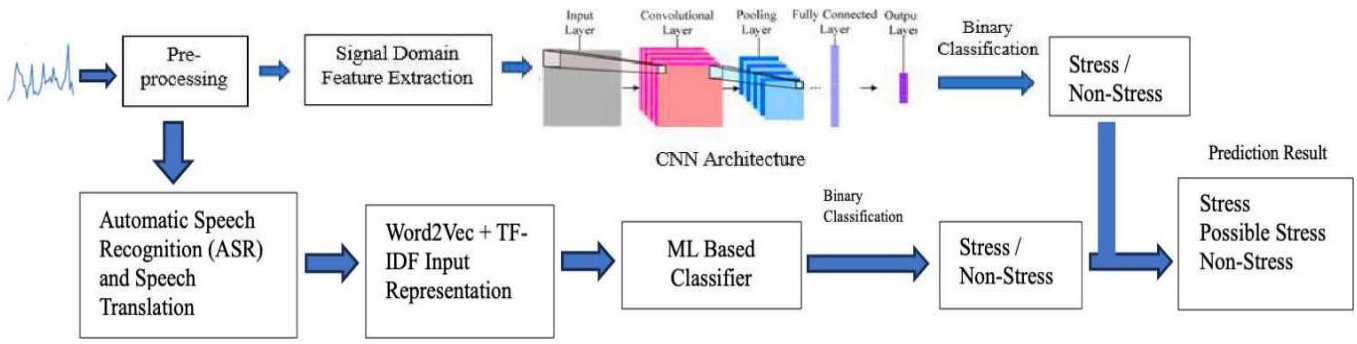

Fig.1. Multi-stage stress detection model

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shifting to add a reverb effect to the original sound sample. This process simulates scenarios encountered in most real-world conditions where voice is acquired. Following the augmentation process, an audio feature extraction process is required so that the dataset can be used for model training. This process aims to translate different audio signal properties into a numerical representation that deep learning algorithms can understand. Mel-Spectrogram audio features in the cepstral domain were used in this study to extract audio features. Due to its ability to perceive sound in a manner similar to the human auditory system, this feature is frequently used in speech recognition and audio analysis applications.

B. Second stage: Stress Detection using Lexical Analysis

The voice recognition process is used in the second stage of the stress detection model to determine the final prediction of the subject's stress status by analyzing the speech's lexical features. Whisper-Medium, an OpenAI-developed pre-trained automatic speech recognition (ASR) model, is used for recognition. Whisper is intended to convert spoken language into text (Speech to Text) and supports multilingual capabilities thanks to the extensive data collection used to train the model. The data pipeline process, which includes data preprocessing, transformation, training, and deployment models, is carried out after the acquired voice has been converted into text. This process is crucial in working with Natural Language Processing (NLP) based models [29]. The completed data pipeline process is briefly shown in Fig. 2.

We used the publicly available open-source dataset Dreaddit, a multi-domain social media text data collection for stress identification, to identify stress conditions through text [30]. As shown in Table I, this dataset includes 3553 posts with binary labels (Stress and Non-Stress) from five general problem domain categories that frequently cause stress, including social problems, abuse, anxiety, PTSD, and financial problems. The average length of posts in the dataset is 420 tokens, with 3,553 labeled segments. We use TF-IDF and Word2Vec for input representation. Word2Vec with TF-IDF is a Word Embedding-based model with statistical weighting and dense vector representations of words, or subwords, which help capture semantic relationships between words.

TABLE I. STATISTIC OF DREADDIT DATASET

Domain	Total Post	*Avg. Token / Post*	*Labeled Segment*
Abuse	2,901	402	703
Anxiety	59,208	191	728
Financial	12,517	198	717
PSTD	4,910	265	711
Social	107,908	578	694
Total	**187,444**	**420**	**3,553**

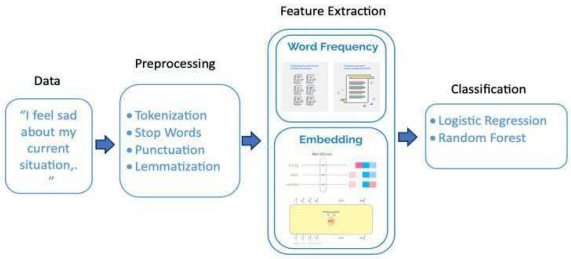

Fig.2.. NLP pipeline for stress detection model

IV. EXPERIMENT AND RESULT

As an experiment from the first stage we used Crema-D and SAVEE datasets to train the speech signal-based stress detection model. After going through a data augmentation process, the dataset increased to 9840 files, consisting of 5520 and 4320, each representing stress and non-stress. To train a CNN-based supervised model with the sequential Conv1D model used, we split the training data and train with a random 75:25 configuration. The CNN architecture consists of 3 convolution layers with a max pooling layer, a flattened layer and two dense layers. For feature extraction, we used the mel-spectrogram and got 131 signal features, which were then used as an input layer with shape (131,1) and produced output (131,256) through 256 channels using a 5x5 filter. The second and third convolutional layers consist of 256 and 128 channels with 5x5 filters, followed by a max-pooling layer. A dense layer with two neuron units with the SoftMax activation function is used in the training process with 50 epochs with a batch size of 4 using the ADAM optimizer. As shown in Fig. 3, the model's accuracy with training data was 0.96 with a loss of 0.05, while for test data it was 0.93 with a loss of 0.16.

In the second stage of the experiment, we trained lexical-based stress detection model with the dreaddit dataset, which contains 3553 data in total, and divided into train and test data with 2883 and 715 data, respectively,. The next step is performing NLP pipeline sequence including tokenization, removing punctuation, removing frequently used words (stopwords), and stemming or lemmatization. Then, Word2Vec is used to extract features with training parameters of vector size=300, window=10, min count=2, and number of workers=10 for 100 iterations. Word embedding, which is a weight vector from hidden layer neurons for each word in the corpus, is obtained after the model has been trained. This weight will establish the semantic relationship between words. Based on the similarity weights assigned to each word, Fig. 4 depicts their semantic relationship. Following word embedding, we weight the words using TF-IDF, and after that, we train the model using random forest and logistic regression classifiers to utilize the lexical features used. The accuracy results from the model training were 70.6% and 69.7% for each classifier. The metric results of each classifier are displayed in Table II.

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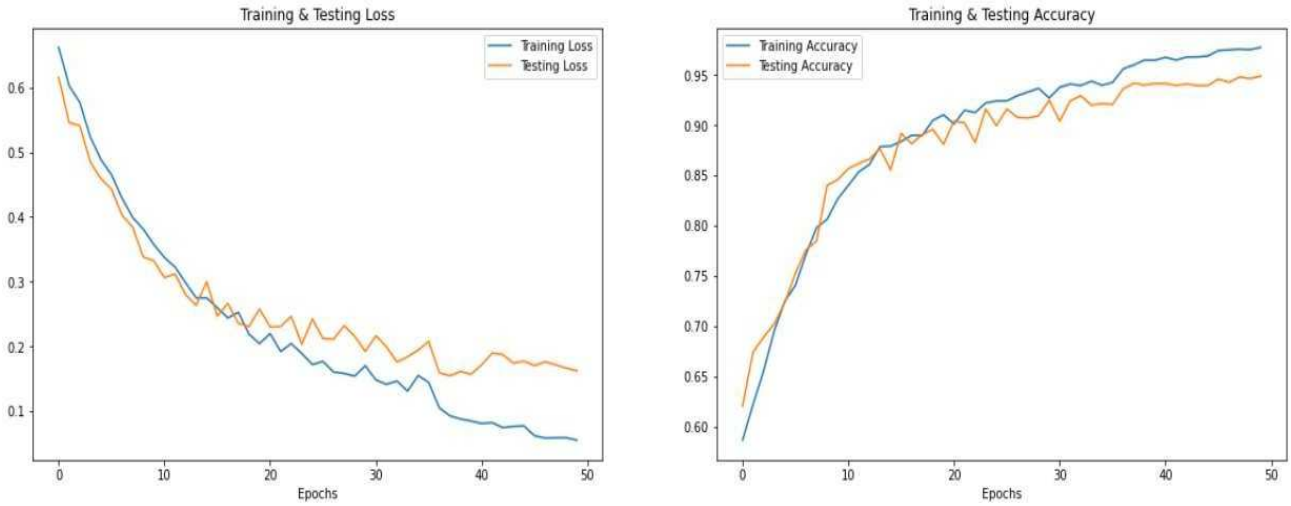

Fig.3. Graph represent model's loss value (left) and accuracy (right)

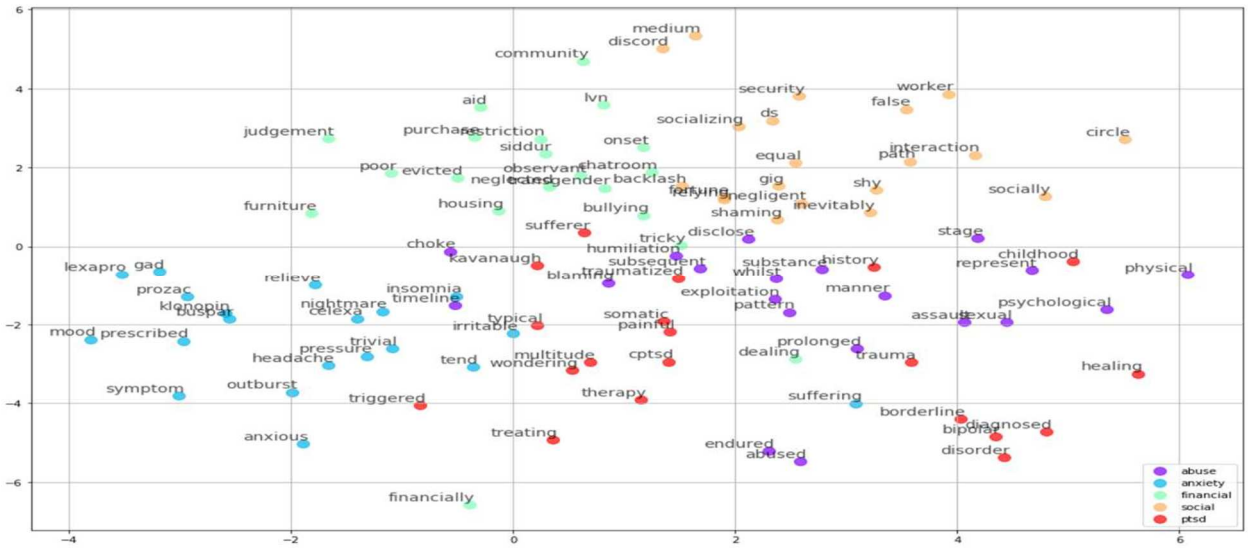

Fig.4. Similar word based on vector space on the dataset

Finally, to conduct a comprehensive test of the stress detection model, we use voice samples from the SAVEE Dataset; we extract speech signal properties to detect stress through speech and also carry out lexical stress detection by capturing the contextual meaning of speech in the voice samples. We used 100 voice samples, each with 50 voice labeled as stress and non-stress, to test our model. Table III shows several samples from the dataset and the prediction results from the model. The prediction results from the model are divided into three categories: Stress, Possible Stress and Non-Stress. In the model performance evaluation, the Stress and Possible Stress categories are counted as Stress. As a result, an average accuracy of 91.7% was obtained, with a precision value of 87.6%, Recall 94% and an average F1-Score of 91.2%. Currently, with an Intel Core i7-10750H based computer test bed with 16GB RAM, the model requires a total processing time of approximately 35 seconds for an audio duration of 5 seconds. So, it is not yet possible to carry out the process in real-time.

TABLE II. CLASSIFICATION PERFORMANCE METRIC OF LEXICAL-BASED STRESS DETECTION MODEL

Classifier	Log Reg (%)	Random Forest (%)
Accuracy	70.6	69.7
Precision	72.0	72.0
Recall	78.3	75.6
F1-Score	75.0	73.2

TABLE III. COMPARISON BETWEEN GROUND TRUTH AND MODEL PREDICTION

Speech Sample From Dataset	Ground Truth	Model prediction
"But the ships are very slow now and we don't get so many sailors anymore."	Stress	Stress
"Project development was proceeding too slowly."	Stress	Stress
"He would not carry a briefcase."	Non-Stress	Possible Stress
"She had your dark suit and greasy washwater all year."	Non-Stress	Non-Stress

V. CONCLUSION

In this paper, we propose a multi-stage stress detection model using a machine learning approach to detect stress through signal characteristics of speech output and combined

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with lexical analysis of spoken sentences. Better generalization capabilities in detecting stress through speech as a signal output and as a linguistic product are provided by our stress detection model, which combines these two methods in a multi-stage manner. Based on the model's evaluation, it produced good results, with an average F1-Score accuracy of 91.7% and 91.2%. In future work, we will concentrate on enhancing the model's performance to reach nearly real-time processing levels, allowing it to detect and monitor the subject's stress condition and making it possible to create a model that is more capable of identifying the subject's stressful moments as well as potential stressors or stress state triggers.

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